

Victor Chua

Singapore | +65 9190 5651 | chua1102@e.ntu.edu.sg | github.com/zeppyclown | linkedin.com/in/vicchua123

Singaporean Citizen | **Available full time from January 2027 for 5 to 6 months**

Computer Vision on Document Images | Dataset Curation & Annotation | Model Training, Evaluation & Reporting

Machine Learning Project: Recognition on Printed Document Images

Drum Hub – optical recognition of printed sheet music | hf.co/Icyautumn/drum-omr-mobilenetv3 Jun 2026 – Present

- **Built the annotated dataset myself** because no open labelled dataset for this notation existed: **19,948 labelled images** from 264 sources across 19 symbol classes and 10 duration classes, split **by source document** rather than at random so no page leaks between train and test
- Ran **20+ rounds of 30-sample random QC** on the labels at 100% agreement before training, and augmented with rotation, brightness and blur so the model holds up on skewed, faded and low-quality scans rather than only on clean renders
- Designed and trained a **dual-head MobileNetV3-Small** (2.47M params) by two-phase transfer learning, classifying symbol type and duration **per position across the page** in one pass: **90.9%** duration accuracy, **97.4%** per-cell accuracy, F1 0.93 and 0.89 on the two most common classes
- Evaluated on **1,924 held-out images from 26 unseen sources** and wrote up the failure modes rather than only the headline number; exported to **ONNX** and published weights under MIT, so inference runs locally at no per-image cost
- Built the reviewing front end in **TypeScript** and **Electron**, rendering recognised output as an editable score with note-level correction, so a human can inspect and fix model output instead of trusting it blind

Work Experience

SUSS Digital Solutions & AI

Singapore

AI & Software Engineer Intern

May 2026 – Aug 2026

- Built an end-to-end automated quality-assessment pipeline over 3-hour lecture recordings: local transcription, per-module scoring against a rubric I designed, composite ranking across the cohort, and a regenerated staff-facing **HTML report and dashboard** on every run
- Made the scoring layer **diagnostic rather than a single number**, surfacing which factor drove each low score so a reviewer sees the likely cause instead of a verdict, and flagging low-engagement segments by timestamp for review
- Replaced LLM-graded scoring with a **deterministic 7-factor formula** over measured acoustics, so every reported figure reconciles to a stored measurement; benchmarked a large model against that formula and kept it out of the scoring path
- Caught that a headline accuracy figure had been measured on the same examples used to ground the judge, re-ran it on **held-out** data, and reported both, in a team codebase where the contaminated number was already circulating

DBS Bank

Singapore

Data Engineer Intern

May 2023 – May 2024

- Built a **Grafana/SQL** dashboard tracking memory and runtime across a pipeline running **3,000+ jobs a day**, broken down per pipeline version so a regression could be attributed to the release that introduced it, which is how teams reached **20% RAM** and **15%** runtime gains
- Wrote a **REST client** for the team's **Elasticsearch** store into the shared internal Python library, giving production models real-time feature retrieval through common infrastructure for the first time

Yap Motor

Singapore

Machine Learning Intern

Dec 2025 – Jan 2026

- Shipped the company's first production AI system, a retrieval service over 500+ pages of multilingual manuals used daily by **24 mechanics** and still running **6 months** later; reached **90% retrieval precision** across 8 diagnostic categories, validated against a month of structured feedback on live queries rather than a static test set

Research

Generative HCI Translation Interface – NTU Turing AI Scholars Programme

May 2025 – Aug 2025

Co-designed a search-based LLM paradigm for computer-assisted translation and solely implemented the bidirectional generation state machine. Research poster presented at NTU TAISP, supervised by Dr Grandee Lee; paper in preparation.

Education

Nanyang Technological University, B.Comp (Computer Science), Turing AI Scholars

Aug 2024 – Dec 2028

Coursework: Machine Learning, Databases, Data Structures & Algorithms, Software Engineering

Temasek Polytechnic, Diploma in Information Technology

2021 – 2024

Technical Skills

ML & Computer Vision: PyTorch, MobileNetV3, transfer learning, multi-head architectures, image classification on document scans, data augmentation, ONNX Runtime, HuggingFace

Dataset & Evaluation: Annotation pipeline design, leakage-safe splitting, sampled QC audits, held-out evaluation, per-class F1, failure-mode analysis, LLM-as-judge rubrics

Interfaces & Dashboarding: Grafana, Streamlit, self-contained HTML reporting, Electron, TypeScript, image and document upload pipelines

Languages & Backend: Python, SQL, TypeScript, JavaScript, FastAPI, PostgreSQL, Elasticsearch, ChromaDB, Git